

Manual

Dynamic Modification of Goods Receipt Inspections WEP-DWP 8.1

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1 Dynamic Modification of Goods Receipt Inspections - Overview

Purpose

This component expands the inspection planning functions and the processing of inspection plans to include dynamic modification of the inspection size based on the batch size and historic inspection decisions.

Implementation Considerations

If the same articles are received repeatedly from the same supplier with a corresponding high batch throughput, use of this component is recommended since dynamic modification can dynamically modify the inspection size and in the case of good quality, can reduce the inspection size.

Integration

This component mainly serves the components "Production Control Inspection Planning" and "Recording and Information Functions for Quality Data".

Features

The following functions are available:

- Use of DIN ISO 2859 and DIN ISO 3951 standards
- Definition of custom inspection severities with their translation rules
- Creation of custom sampling plans
- Choice of characteristic or batch related dynamic modification of inspection sizes
- Evaluation of dynamic modification history with the capability to correct the current inspection severity

2 Inspection Planning

Summary

Menu	Quality management → In-production inspection → Inspection planning Quality management → Goods receipt → Inspection planning Quality management → Goods issue → Inspection planning Quality management → Initial sample → Inspection planning Quality management → Gage management → Inspection planning
Transaction code	iplp
Function authorization	iplp

Utilization

Subject to the field of application, inspection planning is the most important function, as inspection plans are the basis for the generation of inspection orders and the resulting inspections. This means inspections in the area of goods receipt, production and goods issue as well as initial sample inspections and the calibration of test equipment and measurement equipment. As the requirements as regards inspection planning are nearly identical in all areas, the HYDRA inspection planning functions are also nearly identical for these areas.

In creating inspection plans, the user is supported in most cases by the master data catalogs. For this reason it is very important to conscientiously maintain this master data.

Integration

Inspection planning is a prerequisite to generating inspection requirements, the corresponding inspection steps and, as a result, the relevant inspections/calibrations.

The inspection plan in use is referenced in the generated inspection requirements. Moreover, some evaluations/reports also refer to inspection planning. Consequently, data of the same inspection plan number can be added to the basis of a control chart for inspection order characteristics. The same applies to the global control chart analysis, where special inspection plan filter fields may be used. The global control chart analysis does not take into account calibration data and, as a result, no filtered calibration inspection plans.

Another essential relation has been established to the production control plan. As no production control plan can be created without having generated inspection plans beforehand.

Prerequisite

Relevant master data needs to be edited/maintained to be able to create inspection plans. Which master data has to be maintained depends on the respective field of application. However, it is a fundamental prerequisite that articles/items and characteristics are maintained beforehand. Article groups also need to be created, provided that inspection plans based on article groups are created.

Selection criteria

The paragraph that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt and goods issue. By default, the following areas are available.

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage management: Gage management

Active

By checking the corresponding field, the inspection plan list can be restricted to active inspection plans. If this checkbox is not checked the list only shows inspection plans that are in the "in process" and "released" status. The third state of this checkbox (grayed out) shows all inspection plans. This is set by default.

Article number

Filters the article number with respect to inspection plans based on articles.

Article designation

Filters the article name with respect to inspection plans based on articles.

Article group

The article group tree can be opened by clicking the  icon, in case inspection plans for article groups are to be filtered. A function to cancel and accept the entries made is provided.

Operation

Operation number

Customer number

Direct input or opening of the customer catalog and taking over of the number from the entry selected there.

Supplier number

Direct input or opening of the supplier catalog and taking over of the number from the entry selected there.

Manufacturer number

Direct input or opening of the manufacturer catalog and taking over of the number from the entry selected there.

Field descriptions***Inspection plan header*****Area, inspection plan number, inspection plan index**

Uniqueness of all existing inspection plans is secured by the "area", "inspection plan number" and "inspection plan index". The area may be selected. The inspection plan number and inspection plan index may be entered using alphanumeric characters. All three fields are mandatory fields.

The input of these three pieces of information must be unique, i.e. no other inspection plan may exist that already includes this information. By assigning a structured inspection plan number, it is possible to provide specific information. This information might be useful later during sorting. As an alternative, the inspection plan number may also be used synonymously with the article number or gage group number. The inspection plan index corresponds to the inspection plan version. If an existing inspection plan version is to be modified, yet it cannot simply be edited because it has already been used to generate inspection orders, it is recommended to copy the original inspection plan and then modify the inspection plan index (e.g. incrementing it by 1). The inspection plan number should be maintained, if possible, to facilitate later evaluations/reports (e.g. for displaying control charts applying to several orders, in which case the inspection plan versions on which the evaluation is based differ while the inspection plan number is identical).

Inspection plan type

The inspection plan types "article inspection plan" and "group inspection plan" are distinguished. The "article inspection plan" type allows for an article to be assigned from the article master data for which this inspection plan is to be created. The "group inspection plan" option has to be selected to create an inspection plan for article groups. This allows for an article group to be selected. The selected article group is displayed in a separate field within the tree structure.

Article

Input of the article number. Provided that it is known, it may be entered directly. Otherwise, the article catalog can be opened and the requested article can be identified and taken over using the filter and sort criteria provided there. By selecting an article, the drawing issue number, article designation, customer article number and the drawing number are taken over from the master data record and displayed in the corresponding fields.



The customer article number is not displayed within the goods receipt inspection planning.



Nothing may be entered here if the field "article number" or "article group" is shown in gage management.

Drawing issue number

The drawing issue number can be entered directly, just as it is the case for the article number.



In case the article number and the drawing issue number are entered directly, the master data record is determined on the basis of this information while saving and the article designation, customer article number as well as the drawing number are displayed.



The article is uniquely identified by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.



Nothing may be entered here if the "drawing issue number" field is shown in gage management.

Operation assignment

One of the two assignment types "an inspection plan for each OP" and "an inspection plan for all OPs" may be selected.

By assigning operations, it is defined whether the inspection plan to be created should include the characteristics of different operations ("one inspection plan for all OPs") or whether the inspection plan should only include the characteristics of one operation. The fields "operation" and "operation designation" are shown if the "an inspection plan for each OP" option is selected. In any other case, operation details are assigned in the area of inspection plan characteristics. The option "one inspection plan for all OPs" should be selected as far as possible. This allows, for example, printing out of all inspection characteristics of an article, although it has several operations.



The option "one inspection plan for all OP" should be used in areas that are not assigned to a production order. This provides the advantage that the "fictitious" operation only has to be entered once in the inspection plan header, instead for every inspection plan characteristic.

However, the system has to be configured accordingly by MPDV consultants to be able to use this inspection plan variant.

Operation, operation designation

If operations are used, only one of the two fields "operation" or "operation designation" needs to be filled out. Only the operation number has to be entered in the "operation" field if an inspection requirement/inspection order is to be generated automatically by logging an operation of a production order on.

It has to be taken into consideration that these fields are used as search criteria when inspection orders are generated at a later point in time. Provided that only the designation is entered for the operation within the inspection plan, exclusively this information needs to be provided when inspection orders are generated.



A "fictitious" operation has to be assigned (e.g. 9999) even in areas that are not assigned to a production order (e.g. in goods receipt or in gage management). This is necessary to generate inspection steps at a later point in time.

Released/active

Shows whether the inspection plan is "released" and/or "active". If the inspection plan is released or active the corresponding checkboxes are checked. An inspection plan is released and activated, i.e. its status is changed, only by using the corresponding toolbar functions. An inspection plan has to be released before it can be activated.



Only active inspection plans are taken into consideration, when inspection orders are generated at a later point in time.

Released by / on

Shows the HYDRA user who has released it. The release date is displayed additionally.

Valid from / until

Instead of an "unrestricted" activation (using the toolbar), a validity period may be entered here. This period is then taken into account when the inspection order is generated. Yet activation for a certain period means that the user has no clear overview of currently valid inspection plans, and it is therefore recommended to use the "global/unrestricted" activation option using the toolbar if possible. If activated via the toolbar functions, the system carefully monitors whether an active inspection plan exists for the specified article and checks whether this plan includes the same drawing issue number, customers and suppliers. If this is indeed the case, the previously active inspection plan is automatically deactivated.

Type of inspection

Inspection based on characteristics or pieces

When it comes to an inspection based on characteristics and, say a sample size of 5, the characteristic is checked completely first. Whereas with the inspection based on pieces, the characteristic is changed every time a measured value is collected, as items are checked entirely.

Action

Create inspection step or create and immediately release inspection step

Only inspection steps that have been released can be checked.

Customer / supplier / manufacturer

If a customer / manufacturer or supplier is selected this inspection plan only applies for the selected company. Consequently, the customer / manufacturer / supplier becomes the key field of an active inspection plan. In this case, the inspection requirement must also include a company entry in order for this inspection plan to be used for the generation of the inspection requirement.

Dynamic modification type

Characteristic-related, batch-related or none

The authorization "iriscp.dynamic" is required to show these fields.

Batch-related: In this case, it is defined on the level of inspection plan headers which transitional definition is to be used for the dynamic modification. In addition, the initial inspection severity is also defined on the level of inspection plan headers. Now only the reference to the dynamic modification norm is required on the level of inspection plan characteristics. In this context, however, the selection is restricted to those dynamic modification norms referring to the same inspection severity definition as the assigned transitional definition.

Characteristic-related: In this case, the transitional definition, initial inspection severity as well as the dynamic modification norm are referenced on the level of inspection plan characteristics. In this context, however, the inspection severity definition of the assigned transitional definition has to match the assigned dynamic modification norm.

None: The "dynamic modification" function has not been enabled for this inspection plan.

Transitional definition

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is selected. It provides the possible inspection severities and controls switching between the inspection severities.

Initial inspection severity

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is selected. It defines which inspection severity is used for checking the first goods received according to the basics for the dynamic modification history.

Initial sample form

The different form types available for initial sampling can be chosen here. The form type "VDA volume 2, 4th edition" is only supported at the moment. Subject to the selected form type, a corresponding list of categories is shown in the inspection plan characteristics.

Editing functions

The key fields "area", "inspection plan" and "inspection plan index" cannot be changed in the editing mode.

Toolbar



Copy

The below dialog opens to copy an inspection plan:

The target area type and target area may be entered here. Normally, the user should select an area that is identical to that of the source inspection plan. The new inspection plan number and inspection plan index are to be entered afterwards. In case a new version is generated from an existing inspection plan, normally the same inspection plan number is used and only the inspection plan index is changed.

**Activate**

Function authorization: iplp.activate

Makes the inspection plan status "active".

**Deactivate**

Function authorization: iplp.deactive

Puts an inspection plan that is in the "active" status back to the "released" status.

**Release**

Function authorization: iplp.release

Puts an inspection plan that is in the "in process" status to the "released" status.

**In process**

Function authorization: iplp.inprocess

Puts an inspection plan that is in the "released" status in the "in process" status.

"Print" detail application

Function authorization	iplp.print
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The print dialog of the inspection plan header opens a list of available reports. These are Word forms. The potential content of these forms is determined by the Web services that are available in the respective context. The form entries, i.e. the content of the form list of the corresponding print dialog, are defined within the master data of quality management. This is where the basis for new forms is established and the corresponding form properties are defined. A corresponding license is required to be able to change the forms with respect to content and design.

Toolbar - print

There are no other special function buttons in addition to the standard functions.

"Inspection plan characteristics" detail application

The detail application of inspection plan characteristics is nearly identical to the master data of characteristics application. For this reason, reference is made only to additional features in the following.



Go to

For further information on the definition of characteristics, please refer to the functional description of the document entitled [MOC_CharacteristicsQM](#).

On the level of inspection plan characteristics the corresponding inspection plan characteristics are assigned to the previously defined inspection plan header. Assignments are performed by way of initial data creation using the characteristics catalog and taking over the characteristic selected there. By accepting the characteristic, all master data entries are copied to the inspection plan characteristic. Every (copied) detail may be changed and / or complemented afterwards. The characteristic designation is often complemented to define it in more detail.

It is also possible to create a characteristic not included in the characteristics catalog. However, this is recommended only in exceptional cases, since all analyses (e.g. failure mode analysis) are based on characteristics found in the catalog. It is therefore preferable to properly maintain the characteristics catalog.

Different properties and settings can be defined, before specific characteristic data is complemented.

Field descriptions

Inspection plan characteristics

OP sequence

The operation sequence number (OP sequence/AFO) determines the order in which the inspection is later performed. Entries must be unique. In the ideal case, operation sequence numbering should be assigned in steps of 10. This provides the option of later inserting a new characteristic between two existing ones.

Characteristics number

Number of the characteristic selected from master data

Characteristic type

Decides whether the inspection is based on the collection of measured values (variable) or on the specification of the number of detected failures (attributive). When it comes to the attributive inspection, the decision is often only based on the "pass" or "fail" results. Further characteristic types are the inspection chart and the information characteristic. The information characteristic has only been designed to display a document while the inspection is running. Subject to the input type, the lower area of the dialog provides corresponding sampling schemes.

Inspection result base

Decides whether all samples or only the last sample is used to determine the inspection result (pass/fail).

Mandatory inspection

If this checkbox is activated, at least one measured value has to be entered for this characteristic, before an inspection order can be completed with this characteristic.

Calculate characteristic

Identifies a characteristic to be calculated

Formula

Further details can be taken from the manual dealing with the CAQ master data characteristics.

Initial sample creation

Shortlist of categories for initial sampling. The contents depend on the form type selected in the inspection plan header. Within the resulting initial sample report, the inspection plan characteristics are summarized by category and a separate page is printed for each group.

Copy characteristic

It may be defined here which characteristic is to be copied from an initial sample inspection plan to a production plan, for example. Only those characteristics are copied that have been marked relevant in this field.

Details

Defines whether - with respect to this characteristic - the contents of the "details" tab are defined within the inspection plan or whether they have to be determined from the available master data characteristic or an entry from the specification list when inspection requirements are later generated on the basis of this inspection plan. If the "from characteristic catalog" option is selected the "details" tab is hidden. The "from characteristics catalog" option is reasonable if it is a characteristic the specifications of which are to be defined identically for several articles (even applicable for several article groups). In this case, specifications can be defined centrally within the master data. This reduces the input and update efforts considerably.

Specifications

Defines whether - with respect to this characteristic - the contents of the "specification" tab are defined within the inspection plan or whether they have to be determined from the available master data characteristic or from a specification list entry, when inspection requirements are later generated on the basis of this inspection plan. The tabs "specifications", "chart 1", "default values chart 1", "chart 2" and "default values chart 2" are hidden if the option "from list" or "from characteristic catalog" is checked. The "from list" option is only reasonable if inspection plans for article groups are in use. The specification list is used for an inspection plan for article groups if the specifications vary from item to item for certain characteristics.

Dynamically modified

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. This option allows for definitions to be made that a characteristic is not to be modified dynamically, although the dynamic modification option is selected in the inspection plan header.

Transitional definition

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. The transitional definition that can be selected provides the possible inspection severities and controls switching between these different inspection severities. For the dynamic modification based on batches, the transitional definition is assigned in the inspection plan header and is not available on the level of inspection plan characteristics in this case.

Initial inspection severity

The authorization "iriscp.dynamic" is required to show these fields.

It defines which inspection severity is used for checking the first goods received according to the basics for the dynamic modification history. Only the inspection severities of the previously selected transitional definition are available. For the dynamic modification based on batches, the transitional definition is assigned in the inspection plan header and is not available on the level of inspection plan characteristics in this case.

Norm

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. The dynamic modification norm to be used is assigned from master data.

Inspection level

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. An inspection level can only be selected if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859.

AQL

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. An AQL value can only be selected if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859.

Method

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. A method can only be selected if the assigned norm corresponds to DIN ISO 3951.

**Go to**

The fields of the respective tabs correspond to those of the characteristics master data. Further information on these fields can be gathered from the functional description of the document entitled [MOC_CharacteristicsQM](#).

Inspection plan characteristics - toolbar**Copy**

Data of the selected characteristic is opened in insert mode to be able to copy inspection plan characteristics. All fields can be changed. An operation sequence number that does not yet exist in this inspection plan has to be assigned to be able to save it.

The screenshot shows the 'Main page' window for 'Copy'. It contains several tabs: 'Characteristic', 'Properties', 'Details', 'Specifications', 'Chart 1', 'Default values chart 1', 'Chart 2', 'Default values chart 2', and 'User fields'. The 'Characteristic' tab is active, displaying the following fields:

- OP: 10
- Characteristic no.: 0001
- Characteristic designation: Total length
- Machine: ...
- Group: ...
- Planned on: ☒ Machine ☐ Group
- Characteristic type: ☐ attributive ☐ inspection hint ☒ variable
- Inspection result base: ☐ all samples ☐ last sample
- ☐ Mandatory inspection
- ☒ Formula
- Formula: ...
- Inspection station: 03-001
- Operation: 30
- Operation designation: Assembly

"Inspection plan documents" and "documents of inspection plan characteristics" detail applications

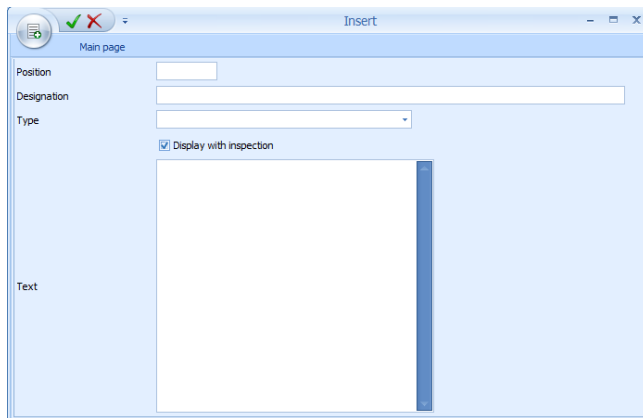
The screenshot shows two windows. The 'Inspection plan' window displays a table with columns: Insp. pla..., Insp. plan index, Article, Drawing issue nu..., Released, Active, Status, and Article designation. The 'Insp. plan document' window displays a form with fields: Position (1), Designation (reference model), Type (file), and a checkbox for 'Display with inspection' (checked).

The above screenshot shows how an inspection plan document is assigned

The screenshot shows two windows. The 'Inspection plan' window displays a table with columns: Insp. pla..., Insp. plan index, Article, Drawing issue nu..., Released, Active, Status, and Article designation. The 'Characteristic document' window displays a form with fields: Position (1), Designation (test example), Type (file), and a checkbox for 'Display with inspection' (checked).

The above screenshot shows how a document of an inspection plan characteristic is assigned

Provided that the "inspection plan documents" tab or the "characteristic documents" tab has been activated in the master detail grid, as many documents as required may be assigned to each inspection plan and each inspection plan characteristic. By enabling these tabs, the toolbar offers corresponding buttons to edit the documents.



When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

The file types are: "FILE", "URL" and "Text". The file name including path may be entered manually with the "file" type. The "URL" file type enables access to the Internet or Intranet. The third file type "Text" allows for text to be entered directly.

A designation may be assigned to each defined document. Moreover, it may be determined in which order the documents are to be listed. The "position" field is used for this purpose (numeric input). The specifications made within this list must be unique. In addition, the checkbox "display with inspection" determines whether or not the document may be shown during the inspection process.

"Inspection plan documents" and "documents of inspection plan characteristics" - toolbar

In addition to the standard features, there is also a button to display the documents.



Show documents

If a document link is defined this button opens and shows this document. However, a program, which is able to visualize the linked file type, has to be installed on the PC.

3 Inspection Severity Definition

Summary

Menu	Master data → Quality management → Inspection severities
Transaction code	isev
Function authorization	isev

Inspection severities represent the single steps of dynamic modification and are an integral part of dynamic modification.

Utilization

The “inspection severity definition” field uniquely identifies inspection severity catalogs within the corresponding QM applications. This field is the key field at the same time, i.e. while saving it is checked whether there is already a data record with this key information.

Different inspection severity catalogs/definitions can be created. They are assigned the individual inspection severities. By default, an inspection severity catalog includes the inspection severities of DIN ISO 2859 or DIN ISO 3951. These are the inspection severities:

- n = normal
- r = reduced
- v = increased and
- a = suspended.

The inspection severity number and designation may be selected individually.

Integration

The following applications use inspection severities.

- Transitional definition (master data – quality management)
- Dynamic modification norm (master data – quality management)
- Inspection planning (inspection plan header and inspection plan characteristics) for goods receipt and goods issue
- Dynamic modification history

Prerequisite

There are no functional requirements that have to be met before inspection severities can be defined. However, this application, if considered individually, is not functional. That is why the applications mentioned in the section entitled "integration" have to be used.

Selection criteria

Selection criteria are self-explanatory and are not described separately.

Field descriptions

The available fields are self-explanatory and are not explained separately.

Toolbar

Inspection severity definition



Assignment of inspection severities to an inspection severity catalog

Function to assign inspection severities to a previously defined and selected inspection severity catalog.

Detail applications

"Inspection severity" detail application

The inspection severity list shows all entries (the actual inspection severities) that are defined for an inspection severity catalog. The detail application of an inspection severity presents the single pieces of information in a clear overview.

Field descriptions

Position

The "position" field defines the order of inspection severities within the list.

Standard

Describes an inspection severity that is to be considered as standard inspection severity.

The other available fields are self-explanatory and are not explained separately.

4 Transitional Definitions

Summary

Menu	Master data → Quality management → Transitional definitions
Transaction code	tdef
Function authorization	tdef

The transitional definitions define according to which rules inspection severities of an inspection severity definition are switched within dynamic modification.

Utilization

The "transitional definition" field uniquely identifies a transitional definition in the corresponding QM applications. This field is the key field at the same time, i.e. while saving it is checked whether there is already a data record with this key information.

The transitional definition "DIN_ISO is an initially existing entry, which can neither be changed nor deleted.

Different transitional definitions can be created. They are, in turn, assigned the transitional rules between the individual inspection severities. Each transitional definition must be assigned a catalog of inspection severities to make sure that a range of inspection severities is available for defining the rules.

Integration

The following applications use transitional definitions.

- Dynamic modification norm (master data – quality management)
- Inspection planning (inspection plan header and inspection plan characteristics) for goods receipt and goods issue

Prerequisite

Inspection severities need to be created before transitional definitions can be defined. The "transitional definition" application, if considered individually, is not functional. To make it a useful application, it has to be used within inspection planning.

Selection criteria

Selection criteria are self-explanatory and are not described separately.

Field descriptions

The available fields are self-explanatory and, as a result, not explained separately.

There is a selection list including acceptance function to assign inspection severity definitions.

Toolbar

Transitional definition



Assignment of rules for switching within inspection severities

Function to assign rules for changing inspection severities to a previously defined and selected transitional definition.

Detail applications

"Transition" detail application

At a glance, the list of transitions shows all entries (the actual rules for the transitions between inspection severities) made for a transitional definition. The detail application for a defined transition makes the single information clearer.

When creating a transition, the inspection severity for which a transition to a reduced inspection severity and to an increased inspection severity is to be created, is to be specified at first. A selection list including the inspection severities is available for this purpose. This list is already filtered to the inspection severity definition that has been defined for the transitional definition.

When switching to a reduced inspection severity, the number of inspections may be indicated that have to be completed with "pass" in a row to go to the new inspection severity. The selection list of inspection severities may be opened in this case as well.

When switching to an increased inspection severity, the number of inspections may be indicated that have to be completed with "fail" in a row to go to the new inspection severity. To do so, the "number of fail inspections in a row" field has to be checked. Once the "number of fail inspections in a row" field has been checked, it may be specified how many inspections out of x inspections have to be completed with "fail" to go to the new inspection severity. In this context, the "fail" inspections do not have to be in a row. The corresponding selection list may be opened here as well to enter the inspection severity.

Field descriptions

In connection with the descriptions made for the "transitions" detail application, the available fields and selection lists are self-explanatory and, as a result, not explained separately.

5 Dynamic Modification Norm

Summary

Menu	Master data → Quality Management → Dynamic modification norm
Transaction code	dynn
Function authorization	dynn

The dynamic modification norm defines how much items/parts are to be checked with which inspection severity taking the batch size into account ("actual quantity" field in the inspection requirement of the goods receipt and goods issue dialog).

Provided that a skip lot variant is to be used, this one is also defined/activated here.

Utilization

The "norm" field uniquely identifies a dynamic modification norm in the corresponding QM applications. This field is the key field at the same time, i.e. when saving it is checked whether there is already a data record with this key information.

The norms "ISO_2859" and "ISO_3951" are initial entries, which can neither be changed nor deleted. Inspection levels, AQLs and sample tables are defined for the norms "ISO 2859" and "ISO 3951" and the methods s and sigma are additionally defined for ISO 3951.

Different dynamic modification norms can be created. An inspection severity definition is assigned to these dynamic modification norms. A selection list that includes the previously defined inspection severities is available for this assignment process.

In addition, a radio button in the "inspection type" field decides whether this norm applies for attributive or variable characteristics only or for attributive and variable characteristics. Consequently, within the inspection plan characteristics of a variable characteristic it is later only possible to choose the dynamic modification norms that have been defined for variable or for variable and attributive characteristics. If the "variable" inspection type is selected the sampling plan type may be indicated additionally. When it comes to the inspection types "attributive" and "attributive + variable", "n-c-d + skip lot" is taken automatically as sampling plan type. The "variable" inspection type additionally provides the option "n-k + skip lot".

n = Sample size

c = Acceptance number (number of errors/defects which is still allowed to achieve a "pass" result)

d = rejection number (number of errors/defects as of which the inspection result is "failed", i.e. the batch is to be rejected)

k = k factor as limit value for the acceptance or rejection / inspection result is classified as "pass" or "fail" (please refer to the corresponding norms for further information on this k factor.)

Integration

The following applications use dynamic modification norms.

- Inspection planning (inspection plan characteristics) for goods receipt and goods issue.

Prerequisite

Inspection severities and transitional definitions need to be created beforehand to be able to define dynamic modification norms. The "dynamic modification norm" application, if considered individually, is not functional. Consequently, it has to be used and assigned in the inspection plan characteristics, which in turn requires the dynamic modification to be activated in the inspection plan header.

Selection criteria

Selection criteria are self-explanatory and are not described separately.

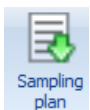
Field descriptions

The available fields are self-explanatory and are not explained separately.

There is a selection list including acceptance functions to assign inspection severity definitions.

Toolbar

Sampling plan



Assignment of sampling plans to dynamic modification norms

Function to assign sampling plans to a previously defined and selected dynamic modification norm.

Detail applications

Detail application sampling plan

The fields "AQL value", "inspection level" and "method" are only available in the initial DIN norms that cannot be changed. The same applies to the sampling plan type "n-k + skip lot". For this reason, the field for the k factor only appears in these sampling plans.

This detail application connects increments of the sample size with a defined batch size for every inspection severity. A selection list is available for the inspection severity.

If an entry is made without specifying the batch size, the specifications of all batch sizes apply that are greater than the highest batch size that is defined.

The copy dialog allows for all sampling plans of the indicated norm and inspection severity to be copied to the specified target norm and target inspection severity.

Field descriptions

Inspection severity

Inspection severity for which the below-mentioned inspection specifications are defined.

Batch size

Gradation for which batch size the below inspection specifications are defined. The indicated batch size represents the upper limit that is included. The lower limit is the next smallest batch size that is defined for the same inspection severity.

Sample size

Defines the number of measured values to be recorded for the combination of inspection severity and batch size.

Acceptance quantity

Limit value for the number of allowed failures/defects with respect to the combination of inspection severity and batch size, which still results in the batch to be accepted or the characteristics to be rated as "pass" for dynamic modification relating to characteristics.

Quantity rejected

Limit value for the number of allowed failures/defects with respect to the combination of inspection severity and batch size, which results in the batch to be rejected or the characteristics to be rated as "fail" for dynamic modification relating to characteristics. Ideally, the rejection quantity is increased by 1 (compared to the acceptance quantity) to be able to make a unique decision.

Skip lot

Provided that a skip lot is to be defined for the specified inspection severity, a value greater than 1 is to be entered here. If the value 3 is entered, this means that only one batch out of three is to be checked as soon as this inspection severity is reached. The sample size as well as the acceptance and rejection numbers that are indicated here apply when checking the inspection severity.